

User Manual

GUI-Based Static and Modal Postprocessing Extension of a 2D Frame-Truss Direct Stiffness Solver

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Project: GUI-Based Static and Modal Postprocessing Extension of a 2D Frame-Truss Direct Stiffness Solver

Course: CE4011 Structural Analysis Software Development

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Python environment used: CE4011, Python 3.11

1. Program overview

This program is a GUI-based educational structural analysis package for two-dimensional frame-truss systems. It supports linear static analysis, modal analysis, model visualization, deformed shapes, internal-force diagrams, result tables, generated frame models, modal mass handling, solver diagnostics, and selected linear modal response-history/response-spectrum postprocessing utilities.

The graphical interface is implemented using Tkinter and Matplotlib. The backend uses a Direct Stiffness Method frame-truss solver and separate modal analysis/postprocessing modules.

The program is not a commercial design package. It does not perform code design checks, nonlinear analysis, full 3D frame analysis, shell/slab modeling, or rigid diaphragm modeling.

2. Main GUI workflow

Typical workflow:

1. Open or define a model.
2. Inspect the model view.
3. Run static analysis.
4. View static deformed shape, reactions, member forces, and N/V/M diagrams.
5. Run modal analysis if modal masses are available.
6. View mode shapes, frequencies, periods, modal participation, and effective modal masses.
7. Export tables or use generated plots for reporting.

3. Main menus

File

Used for model loading, saving, generated examples, and package-style file workflows.

Common actions:

- Open JSON/XML/text model.
- Open generated model.
- Save model.
- Save model as another file.
- Save model package, if available.

When a model uses a companion modal mass file, keep that file together with the main model file.

Define

Used for form-based model editing. Typical definitions include:

- Nodes.
- Materials.
- Sections.
- Elements.
- Supports.
- Nodal loads.
- Member loads.
- Thermal loads.
- Support settlements.
- Modal masses.
- Axis offsets / rigid end offsets.
- Model units.
- Diagram display conventions.

Analyze

Runs the computational modules:

- Run Static Analysis.
- Run Modal Analysis.
- Optional modal response-history or response-spectrum utilities, where available.

Display

Shows graphical outputs:

- Model View.
- Static Deformed Shape.
- Hermite Deformed Shape.
- Axial Force Diagram.
- Shear Force Diagram.
- Bending Moment Diagram.
- Section Force Stations.
- Deformed Slope Profile.
- Mode Shapes.
- Story Drift Profile.
- Roof Displacement / floor response plots.
- RHA/RSA plots, where available.

Tables

Shows numerical results:

- Nodal Displacements.
- Support Reactions.
- Member-End Forces.
- Element Station Forces.
- Element Deformed Slopes.
- Modal Frequencies.
- Modal Participation Factors.
- Modal Response Factors.
- Roof Displacement.
- Story Drift.
- Solver Diagnostics.

Most result tables can be exported as CSV files.

Select

Provides node/element selection tools for inspection. Selection is useful for larger generated frame models.

Help

Provides summary help, model instructions, or workflow notes where implemented.

4. Input routes

The program supports five model input routes:

1. JSON files.
2. XML files.
3. Text-deck files.
4. Form-based GUI input.
5. Generated parametric frame models.

All input routes are converted into the same backend model dictionary before a Structure object is created. This means that a generated model, a text-deck model, and a GUI-defined model use the same solver path after loading.

5. Units

The default convention is:

```
Force: N
Length: m
Mass: kg
Time: s
Temperature: C
Rotation: rad
```

Unit settings are used as metadata and labels in the GUI, tables, and plots. They do not automatically convert existing numerical input values. Users must enter consistent numerical values.

6. Opening a generated frame model

Example workflow:

1. Launch the GUI.
2. Use `File -> Open Generated Model` or the generated-model loading button.
3. Select `model\_a\_5story\_unbraced.json`.
4. Confirm that the model appears in the Model View.
5. Confirm that the summary panel lists the correct number of nodes and elements.

The five-story unbraced frame has five stories, one bay, fixed bases, lateral nodal loads, and companion modal masses. It is used as the main static/modal demonstration case.

7. Running static analysis

1. Load or define a stable model.
2. Inspect geometry, supports, and loads in Model View.
3. Click `Run Static Analysis` or use `Analyze -> Run Static Analysis`.
4. Check the summary panel for analysis status.
5. Open result plots and tables.

Common static outputs:

- Nodal displacement table.
- Support reaction table.
- Member-end force table.
- Deformed shape plot.
- Axial/shear/bending force diagrams.
- Element station-force table.
- Story drift and roof displacement for multi-story frames.

The deformed-shape plot uses visual magnification. Actual displacement values are reported in the tables and summary panel.

8. Internal-force diagrams

The program can display:

- Axial Force Diagram.
- Shear Force Diagram.
- Bending Moment Diagram.

The diagrams are plotted on the actual 2D member geometry using member tangents and normals. Diagram display conventions control the plotted side/sign for visual comparison with tools such as Ftool. Changing the display convention does not change numerical member-force values.

For detailed section results, use Tables -> Element Station Forces.

9. Member loads

Supported member-load inputs include:

- Frame-element point loads in local transverse direction.

- Full-span uniformly distributed loads.
- Supported local axial load effects where implemented by the backend.

Partial UDLs are not part of the final supported core unless explicitly implemented. Unsupported truss/member-load combinations are rejected.

10. Thermal loads and support settlements

Thermal loads and support settlements are inherited from Assignment 4 and exposed through the final workflow. They can be loaded from files or defined through the GUI where supported.

Thermal/settlement effects are included in static analysis. They are not modal masses and should not be confused with dynamic mass input.

11. Running modal analysis

Modal analysis requires modal mass data. Modal masses may come from:

- explicit model entries;
- companion mass files;
- generated-model mass definitions;
- GUI-defined modal mass records.

Workflow:

1. Open a model with modal masses.
2. Click `Run Modal Analysis` or use `Analyze -> Run Modal Analysis`.
3. Select the number of modes to display.
4. View mode shapes and modal tables.

Common modal outputs:

- Circular frequency, ω , in rad/s.
- Frequency, f , in Hz.
- Period, T , in seconds.
- Mode shapes.
- Modal participation factors.
- Effective modal masses.
- Modal mass ratios and cumulative mass ratios.
- DOF classification information for massive/massless DOFs.

Mode-shape sign is arbitrary. If a mode shape appears inverted relative to another program, this is not a numerical error.

12. Companion modal mass files

Some generated models use a separate companion mass file. For example:

```
model_a_5story_unbraced.json
model_a_5story_unbraced_masses.json
```

Keep the mass file in the same expected location. If the companion file is missing, modal analysis may fail or use documented fallback behavior depending on the model workflow. Users should verify that the modal mass source is

physically appropriate before interpreting modal results.

13. Modal response-history and response-spectrum utilities

The program includes selected educational linear modal dynamic utilities:

- Ground-motion record loading.
- Elastic response-spectrum generation.
- Rayleigh damping diagnostics.
- Linear modal response-history analysis.
- Response-spectrum analysis with ABSSUM, SRSS, and CQC combination.

These are modal postprocessing extensions. They are not nonlinear response-history analysis and are not full commercial seismic design workflows.

14. Solver diagnostics

The solver diagnostics table reports computational information such as:

- number of nodes and elements;
- total/free/restrained DOFs;
- active stiffness matrix size;
- nonzero stiffness entries;
- matrix density;
- semi-bandwidth and full bandwidth;
- optional RCM-estimated bandwidth.

This connects the final project back to Assignment 1 topics such as DOF numbering, sparsity, and bandwidth.

15. Saving models

Use the File menu to save the current model. Recommended practice:

- Save the main model file.
- If explicit modal masses are used, save or preserve the companion mass file.
- Do not treat fallback/default masses as physical masses unless they were intentionally defined.

After saving, reopen the model once to confirm that geometry, loads, supports, and modal masses are preserved.

16. Recommended demonstration workflow

For the final project demonstration:

1. Open `model\_a\_5story\_unbraced.json`.
2. Show Model View.
3. Run Static Analysis.
4. Show Static Deformed Shape.
5. Show Bending Moment Diagram or station forces.
6. Open Nodal Displacements and Support Reactions tables.

7. Run Modal Analysis.
8. Show Mode 1 and Modal Frequencies.
9. Show Modal Participation Factors.
10. Show Story Drift or Roof Displacement.
11. Show Solver Diagnostics.
12. Show terminal verification output if needed.

17. Verification examples included with the project

The project includes or supports:

- Cantilever beam static verification.
- Simply supported beam static verification.
- CE586 modal benchmark examples.
- Five-story generated frame validation.
- Autodesk Robot modal comparison.
- Ten-story braced/unbraced generated-frame comparison.
- RHA/RSA supplementary examples.

18. Common warnings and interpretation notes

- A visually large deformed shape does not mean large physical displacement; the plot is magnified.
- Diagram sign/side may differ from Ftool depending on display convention.
- Unit metadata does not convert numbers.
- Modal shape signs are arbitrary.
- Missing modal mass data prevents meaningful modal analysis.
- RHA/RSA utilities are educational modal postprocessing tools.

19. Limitations

The software is limited to linear elastic 2D frame-truss analysis. It does not include:

- full 3D frame analysis;
- shell or slab elements;
- rigid diaphragm constraints;
- nonlinear material behavior;
- geometric nonlinearity or P-Delta;
- staged construction;
- design-code checks;
- complete commercial load-combination workflows;
- nonlinear response-history analysis.

Future extensions may include 3D frame elements, diaphragm constraints, shell/slab elements, partial-span member-load formulas, nonlinear analysis, and code-based design checks.